

▣ 08 Independence

Example

Consider a car presenting for an NCT (national car test – the Irish road worthiness examination for motor vehicles) and the following events.

A: The car passes the test

B: The car is green

C: The car is at least 20 years old.

If we are interested in calculating $\Pr(A)$, we would reasonably assume whether or not event B has occurred should not affect this probability.

I.e., The probability of passing the NCT does not depend on what colour it is.

Or using probability notation: $\Pr(A | B) = \Pr(A)$

We say that the events A and B are not independent. Knowing that B occurs does not change the probability of A occurring,

However, we would likely see events A and C differently. Assuming a higher failure rate for 20-year-old cars we would reduce our estimate of the probability of the car passing if it is at least 20 years old.

I.e., $\Pr(A | C) \neq \Pr(A)$

Events A and C are not independent!

Examples

The event that an individual is a smoker and the event that they develop serious health problems are **not** independent.

The event that the first ball drawn in this week's lotto is 13 and the event that first ball drawn in next week's lotto is also 13 **are** independent.

Example

A drum contains the numbers 1 to 10. One ball is randomly selected from the drum, replaced and then a second is drawn. Define the following events.

A: first ball has a value of '5' or less.

B: first ball has a value of '3' or more.

C: second ball is one of {2, 3, 4}

It is left as an exercise to show $\Pr(A) = 5/10 = 0.5$, $\Pr(B) = 8/10 = 0.8$ and $\Pr(C) = 3/10 = 0.3$

$\Pr(A \cap B) = \Pr(\text{"First ball is 5 or less AND 3 or more"}) = \Pr(\text{"First ball is 3, 4 or 5"}) = 3/10 = 0.3$

What about $\Pr(B|A)$?

Using Bayes theorem $\Pr(B|A) = \Pr(B \cap A) / \Pr(A) = 0.3/0.5 = 0.6$

So, $\Pr(B|A)$ is not the same as $\Pr(B)$

Thus, the events A and B are not independent. Knowing that A occurs changes the probability of B occurring,

What about events A and C?

As the ball is replaced, whether or not A occurs does not affect the probability of C

I.e., Or $\Pr(C|A) = \Pr(C)$. So, A and C are independent.

Simple multiplication law of probability

If two events A and B are independent, then $\Pr(A \cap B) = \Pr(A) \Pr(B)$

Problem 1

How likely is it that a fair coin, if tossed 4 times, will result in tails each time?

✂ *A common error in probability is to assume events are independent when they are not.*

Example (not independent)

We might estimate the probability of a car tyre puncturing on a journey to be 1 in a 1,000 or 0.001. It would be questionable to assume that the probability of two tyres puncturing would be $(0.001)^2$ (1 in a 10^6 !). I.e., a pothole, or debris on the road that might cause one tyre to puncture might also cause a second one to do so.

Example (not independent)

If it is estimated that 10% of the population suffer from a particular medical condition, then it would similarly be questionable to assume that the likelihood of two siblings having this condition would be $0.1^2 = 0.01 = 1\%$. I.e., siblings would have commonality with their genes and with their environment.

Example (independent)

If a fair coin is flipped 5 times and “heads” result on each of the first four, then the probability of heads on the fifth flip remains 0.5.

Example (independent)

Or if, in a lotto draw, a particular number has not been drawn for a long time then it is somehow “due” or that the “law of averages” dictates that it should appear.

Problem 2

A family have two children. If one is a boy, how likely is it that the other is a girl?

The Prosecutors Fallacy

Problem 3

A DNA sample is taken from a crime scene and is found to match an individual. There is almost no other evidence against this individual and without the DNA match, it is (subjectively) estimated that the likelihood of him being the perpetrator of the crime is one in two million. It is further estimate that the chance of the DNA collected coming from someone other than the perpetrator is one in a million. Is it likely that the individual whose DNA matched the collected sample is the perpetrator?

As we will see (in class) the individual is twice as likely to be innocent as guilty!

The above calculations would suggest that DNA evidence of little use. What is being implicitly assumed above is that the individual was probably asked to provide a DNA sample because the probability of guilt was likely much higher than one in two million.

Suppose the probability of guilt without the DNA evidence was only 0.1% then with it, this increases to 0.999

What about the claim that there is only a chance in a million of an innocent person having a DNA profile that matches the crime scene sample? Is this a credible probability estimate?

Yes, if only a single person is compared

But what if the match is made by trawling a DNA database?

Problem 4

The chances of a randomly selected person have a DNA profile matching a specific sample is 1 in one million. How likely is such a match if a database of 500,000 million samples (excluding the person from whom the sample came) is searched?

Notes

The probability of an event being 0.9999 means there is one chance in 10,000 of the event not happening. But the probability of an event being 0.99999 means there is one chance in 100,000 of the event not happening, which in an obvious sense, is 10 times less likely!

✍ *Be careful rounding probabilities that are very close to 1*

Exercise

500,000 people pay a lotto game each week for which the chances of winning or sharing in the top prize is estimated to be one chance in 3.6 million. How likely is it that the top prize will “roll over” three times?

Problem 5

Consider a problem where a playing card is selected at random from a pack of 52 and define the events.

A: Card is a diamond, B: Card is a 4, 5 or 6, C: Card is less than or equal to 4 and D: card is red

Decide if (i) A and B and (ii) B and C are independent.

Decide if (i) A & C and (ii) A & D are independent

Note that demonstrating that $\Pr(B) = \Pr(B | A)$ also establishes that A and B are independent.

Problem 6

A technician is charged with resolving the various IT problems encountered by clerical staff in a utility company. Each problem is classified as hardware, software, network or other and each is classified as having being resolved the same day or not. The problem log for one week shows that the frequency of problem types (the number resolved within one day) are hardware 10 (6), software 16 (12), network 16 (8) and other 8 (4). A entry from the log is randomly selected and the event H denotes that this is a hardware problem. S, N and O are similarly defined and additionally, D denotes the event that the problem was resolved within one day. Decide if (i) R and S, (ii) R and H and (iii) R and N are independent events.

Exercise

A healthcare facility has 20 doctors (9 male), 20 nurses (5 male), 16 administrators (8 male) and 24 other staff members (18 male). For a randomly selected staff member, the events M, A and N are defined respectively to be that they are male, part of the admin staff and part of the nursing staff, show that M and A are independent but N and A are not.

Exercise

An employee is to be randomly designated as being a team member on two of four projects undertaken by a firm. Half of the projects involve working remotely, the rest on-site. Define the events O: at least one assigned project is remote, F: the first project is remote and B: both projects are remote, decide if the events F and B and the events F and O are independent or not.